

TENAGA ENERGY SOLUTIONS LLP
Reg ofc :845-5 Ground floor MKK Building
Near Al Salama Eye Hospital , Perinthalmanna
Pin 679322; Ph 9544243300, 9846243300
Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

Dear sir/madam,

We, at TENAGA ENERGY SOLUTIONS LLP, take great pride in introducing ourselves as a distinguished solar integrator empanelled by ANERT (Agency for New and Renewable Energy Research & Technology), operating under the aegis of the Government of Kerala. ANERT, recognized as the Nodal Agency for the Ministry of New and Renewable Energy Sources (MNRE) at the national level, further underscores our commitment to excellence and innovation in the renewable energy sector.

Accreditations and Authorizations:

Our commitment to quality and sustainable energy solutions is reflected in our empanelment under the DBT rooftop subsidy scheme by MNRE through the Kerala State Electricity Board (KSEBL). **Additionally, we are honored to be an authorized channel partner for Havells solar division.**

Geographical Presence:

With a strategic presence in the Malabar region, our registered office in Malappuram District, a branch office in Thiruvanthapuram & Kasaragod districts, and a channel partnership in Ernakulam district exemplify our commitment to providing accessible and efficient services throughout the region.

Scope of Services:

TENAGA ENERGY SOLUTIONS LLP stands as a dynamic force in the renewable energy sector, offering comprehensive services from conceptualization to engineering design in Renewable Energy Solutions, with a specific focus on solar energy. Our unwavering commitment to quality and efficiency serves as the cornerstone of our operations.

1

Branch offices : Thampanoor , Thiruvanthapuram & Uppala, Kasaragod
Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanthapuram



TENAGA ENERGY SOLUTIONS LLP
Reg ofc :845-5 Ground floor MKK Building
Near Al Salama Eye Hospital , Perinthalmanna
Pin 679322; Ph 9544243300, 9846243300
Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

Commitment to Renewable Energy:

We are dedicated to harnessing renewable energy resources to shape a sustainable and seamless future. Our product development team is steadfast in providing cutting-edge service options that align with the evolving needs of our clients.

Technical Expertise:

Our dedicated team, led by a cadre of skilled engineers and bolstered by the expertise of seasoned solar technicians, stands poised to deliver unparalleled excellence in every project we undertake. Ensuring expertise in on-grid , hybrid and off-grid solar power plant installations. This technical prowess positions us as a reliable partner for your solar energy initiatives.

Quotation Submission:

Enclosed herewith, please find a detailed offer or quotation tailored to your project requirements. We believe our proposal aligns seamlessly with your goals and requirements.

We eagerly anticipate the opportunity to contribute to and collaborate on your esteemed ON GRID SOLAR project. Thank you for considering TENAGA ENERGY SOLUTIONS LLP as your trusted partner in renewable energy solutions.

Please feel free to contact us at 9544243300 for any further clarifications or discussions.
Sincerely,

Nithin MV
Designated Partner
TENAGA ENERGY SOLUTIONS LLP
9544243300

2

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvananthapuram



TENAGA ENERGY SOLUTIONS LLP
 Reg ofc :845-5 Ground floor MKK Building
 Near Al Salama Eye Hospital , Perinthalmanna
 Pin 679322; Ph 9544243300, 9846243300
 Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

TO,

Mr Harilal, Thottinakkare , Mampad

Sub : QUOTATION FOR 5 KW ON GRID SOLAR POWER PLANT (PMSG SUBSIDY)

We are pleased to Quote a best specification as below.

REF:Mr Jafar

<u>S L N O</u>	<u>SYSTEM COMPONENT</u>	<u>CAPACITY/RA TING</u>	<u>MINIMUM TECHNICAL COMPLIANCE</u>
1	SOLAR PANEL: RENEW/PAHAL BIFACIAL PERC EMMVEE TOPCON	RENEW/PAHAL : 9 panels of 545/550 Wp HALF CUT bifacial PV MODULES EMMVEE 565 TOPCON(PRICE CHANGES GIVEN BELOW)	IEC61215/ISI4286:IEC61730 part 1 &2:IEC61701
2	On-GRID Inverter MAKE: SOLIS (1 PHASE) /HAVELLS	5 KW / 1 P-N /2 MPPT/2 string	IEC61683/IS61683(Efficiency 98.7% and above) IEC60068-2(1,2,14,30)/Equivalent BIS Std IEC62116-for islanding prevention compliance to CEA(Distribution) regulation,2013,IEC61727 for utility interface
3	Cables	As per site use wiring length: AC SIDE: 4 sqmm copper	IEC60227/IS694 IEC605502/IS1554(PT.1 &2)

3

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
 Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanthapuram



TENAGA ENERGY SOLUTIONS LLP
 Reg ofc :845-5 Ground floor MKK Building
 Near Al Salama Eye Hospital , Perinthalmanna
 Pin 679322; Ph 9544243300, 9846243300
 Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

		wire/ UG 4SQ MM ; & DC SIDE: Polycab 4 SQ MM(type 1)	
4	AC DB -1 in 1 out (WITH SPD)-1 SET SINGLE PHASE	Switches/Circuit MCB:HAVELLS /VGUARD; SPD:PHOENIX CONTACT/ HAVELLS	IEC60947 PART I &II,III/ IS 60947 PART I,II,III EN 50521
5	DCDB-2 in 2 out WITH FUSE AND SPD-1 SET	MCB:HAVELLS/IN FI/VGUARD; SPD:PHOENIX CONTACT/VGAUR D FUSE:VGUARD/SU NSEED	
6	MC4 CONNECTORS	T2 CONNECTORS NINGBO	
7	Electrical Grounding (EATRHING)	AC DB , DC DB AND LA- NICKEL SEALED COPPER BONDED ROD 14 MM DIA X 1.2 MTR LONG X 100 MICRONS WITH t STRIPS & EARTH PIT CHAMBER-3, MINERALS ETC; COPPER 10SWG / 16 sqmm stranded aluminum down conductor	As per IS 3043: ; KSEB& EI REGULATIONS
8	PANEL SIDE(DC) EARTHING	INDIVIDUAL SOLAR PANELS AND MODULE MOUNTING STRUCTURE EARTHING –	As per IS 3043: ; KSEB& EI REGULATIONS

4

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
 Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanthapuram



TENAGA ENERGY SOLUTIONS LLP
 Reg ofc :845-5 Ground floor MKK Building
 Near Al Salama Eye Hospital , Perinthalmanna
 Pin 679322; Ph 9544243300, 9846243300
 Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

9	MODULE MOUNTING STRUCTURE	ELEVATED BASE STRUCTURE OVER THE SLOPE AND FLAT ROOF OVER THE SOUTH WEST SIDE OF THE HOUSE	AS PER ANERT & MNRE GUIDELINES (This process including panel flooring on module mounting structure) ; **ANY CHANGES ATTRACTS ADDITIONAL COST
10	LIGHTNING ARRESTOR & ALUMINIUM DOWN CONDUTOR	LA: NICKEL SEALED COPPER BONDED -20MM DIA X 1 MTR LONG (100 MICRONS) Single SPIKE WITH INSULATED BASE	As per IS 3043: EXCEL EARTHINGS ,
11	Energy meters	Single directional meter for recording solar energy produced -3PH	As per CEA regulation
A	PROJECT COST		INR 3,05,000
B	KSEB FEASIBILITY, REGISTRATION amount and PAPER WORKS & INSPECTION -5KW		INR 7,080 (Rs 4000 WILL BE REFUNDED BY KSEBL AFTER 6 MONTHS OF PLANT COMMISSIONING)
	2 MTR ELEVATED STRUCTURE WITH GP-16 GRADE SQUARE PIPES; WALKWAY AND LADDER		INR 30,000
	TOTAL PROJECT COST		INR 3,42,080
	SPECIAL DISCOUNT		INR 7,080
	FINAL COST AFTER DISCOUNT WITH RENEW/PAHAL		INR 3,35,000
	SAME WITH EMMVEE TOPCON		INR 3,40,000

5

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
 Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanthapuram



TENAGA ENERGY SOLUTIONS LLP
 Reg ofc :845-5 Ground floor MKK Building
 Near Al Salama Eye Hospital , Perinthalmanna
 Pin 679322; Ph 9544243300, 9846243300
 Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

NOTE:

5 KW on grid plant can produce 20 (AVG) units of electricity per day (provided that uninterrupted KSEB supply and good sunlight availability); MAX PRODUCTION PER DAY 25 UNITS. PLANT GOT FUTURE SCOPE OF 6.5 KW

After 6 months from the date of successful commissioning of solar plant KSEBL will give you a refund of registration amount (excluding GST) approximately INR 4000

NET METER COST NOT INCLUDED USUALLY SUPPLIED BY KSEBL, IN CASE ANY WE CAN SUPPLY THE TESTED NET METER WITH AN ADDITIONAL COST OF INR 4000 FOR VISIONTEK OR INR 5000 FOR L&T (SINGLE PHASE)

TERMS & CONDITIONS:

1)WARRANTY TERMS: AS PER COMPANY POLICIES

ONGRID INVERTER	SOLIS: 10 YEARS OF REPLACEMENT SERVICE WARRANTY AND CAN BE EXTENDED UPTO 20 YEARS WITH ADDITIONAL PAYEMENT HAVELLS : 10 YEARS OF ON SITE REPLACEMENT WARRANTY
SOLAR PANELS	12 YEARS OF PRODCUT WARRANTY & 30 YEARS OF LINEAR PERFORMANCE WARRANTY (10% PERFORMANCE REDUCTION IN FIRST 10 YEARS & 20% REDCUTION AFTER 20 YEARS) BY RENEW/PAHAL/EMMVEE
SYSTEM WARRANTY & PLANT MAINTENANCE	FREE FOR FIRST 5 YEARS: 2 FREE SITE VISIT AND PLANT MAINTENANCE PER YEAR BY TENAGA

6

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
 Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanathapuram



TENAGA ENERGY SOLUTIONS LLP
 Reg ofc :845-5 Ground floor MKK Building
 Near Al Salama Eye Hospital , Perinthalmanna
 Pin 679322; Ph 9544243300, 9846243300
 Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

2)PAYEMENT TERMS:

ADVANCE PAYEMENT	INR 1,00,000
ON DELIVERY OF MATERIALS	
COMPLETION OF MODULE STRUCTURE WORK AND PANEL FLOORING	INR 1,00,000
COMPLETION OF AC & DC SIDE WIRING AND INSTALLTION OF INVERTER & CALIBRATION	INR 75,000
AFTER NET METER ALLOCATION and GRID CONNECTION	BALANCE AMOUNT

BANK DETAILS FOR PAYEMENT:

TENAGA ENERGY SOLUTIONS LLP

AC NO : 0329073000000765

IFSC :SIBL0000329

BANK : SOUTH INDIAN BANK

BRANCH : PERINTHALMANNA

7

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
 Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanthapuram



TENAGA ENERGY SOLUTIONS LLP
 Reg ofc :845-5 Ground floor MKK Building
 Near Al Salama Eye Hospital , Perinthalmanna
 Pin 679322; Ph 9544243300, 9846243300
 Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

3)WORK SCHEDULE:

SCOPE	WORK SCHEDULE DETAILS	TIME LINE
TENAGA	FEASIBILITY REPORT : THOUGH MNRE NATIONAL ROOFTOP SOLAR PORTAL	WEEK 1 (after advance payment)
	DELIVERY OF MATERIALS	WEEK1
	SOLAR PANEL MOUNTING STRUCTURE AND PANEL FLOORING	WEEK 2
	DC & AC SIDE WIRING, EARTHING, INSTALLTION AND INVERTER CALIBRATION	WEEK 3
	SUBMISSION OF COMPLETION REPORT AND CHECKLIST TO KSEB; PLANT REGISTRATION	WEEK 3
KSEB	NET METER ALLOCATION	AFETR KSEBL TESTING AND INSPECTON ; ANOTHER 1-2 WEEKS
	REPLACEMENT OF EXISTING METER WITH BIDIRECTIONAL ENERGY METER (NET METER) AND CONNECTION TO GRID	
SUBSIDY	NEED TO GENEARTE SUBSIDY REQUEST AFTER PLANT COMMISSIOING	60 -90 DAYS AFTER SUBSIDY REQUEST

General Conditions of the System :

- Minimum shadow free space of 100 Sq. Ft or 10 Sq. Mts per KWP of solar capacity shall be available on the proposed area to accommodate solar panels, inverter(s), AJB, ACDB and other vital plant components.
- The system will generate electricity during peak sun hours (day time) only and it can be utilized at the same time or exported to the grid if net-

8

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
 Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanthapuram



TENAGA ENERGY SOLUTIONS LLP
Reg ofc :845-5 Ground floor MKK Building
Near Al Salama Eye Hospital , Perinthalmanna
Pin 679322; Ph 9544243300, 9846243300
Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

metering is enabled. The grid-tied inverter would shut itself down during grid failure in order to comply with the anti-islanding law and it can be synchronized with customer DG set during that time

- The grid-tied inverter will mix solar along with grid power and reduce your electricity bill by getting consumed first by your electrical loads and balance shall be drawn from grid. Based on the generic data, a grid-tied solar system can generate 3.5 to 4 KWH per day per KWP of solar approximately on an average of 365 days

NOTE:

- This quotation is valid only for 10 days
- Quotation subjected to the market changes after the prescribed time.
- Quoted prices are inclusive of GST
- Above quoted prices are inclusive of KSEB & electrical inspectorate charges and paper works

Hope you will find our quote most reasonable and according to your requirement.

Awaiting your valuable purchase order....

NITHIN MV
9544 24 33 00
For TENAGA ENERGY SOLUTIONS LLP

TENAGA ENERGY SOLUTIONS LLP
Reg ofc :845-5 Ground floor MKK Building
Near Al Salama Eye Hospital , Perinthalmanna
Pin 679322; Ph 9544243300, 9846243300
Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

PROJECT ONGOING:

- 1) 120 KW, KANDARIKKAL RUBBERS, PALOORKOTTA ANGADIPURAM UNDER MSMSE SCHEME**
- 2) 50KW IMAGE MOBILES MANJERI, UNDER MSME SCHEME**

SOME OF THE COMPLETED SITE DETAILS :

MORE THAN 1MW OF RESIDENTIAL PROJECTS ALL OVER KERALA AND PART OF KARNATAKA

1. 70KW ONGRID COMMERCIAL PLANT, MALABAR FOODS, CHEMMANIYOD , NEAR MELATTUR , ADANI BIFACIAL MODULES AND SUNGROW STRING INVERTER
2. 45KW ONGRID PLANT IN UNITED LIGHTS, PERINTHALMANNA ; HAVELLS 20KW MI COMBO AND 25KW VGUARD MI COMBO
3. 40KW ONGRID PLANT WITH RENEWSYS TOPCON AND GROWATT STRING INVERTER AT V- CARE HEALTHCARE, PHASE 2 PERINTHALMANNA
4. 30KW ONGRID PLANT VCARE HEALTH CARE OOTY ROAD PERINTHALMANNA , MI COMBO KIRLOSKAR
5. 35 KW ONGRID PLANT, KM REXIN PERINTHALMANNA WITH ENPHASE MICRO INVERTERS AND VIKRAM HALFCUT PANELS
6. 20KW SOLAR ONGRID COMMERCIAL POWER PLANT: CLIENT NAME: KOTTARAM HYPERMARKET ALANELLUR;INVERTER : SOLIS 20KW; P V MODULES : HHV SOLAR MONOPERC
7. 20KW COMMERCIAL PLANT : JANATHA HOME APPLIANCES PERINTHALMANNA ; PV MODULES 540Wp HALFCUT MONOPERC MODULES PF HHV SOLAR, 20KW SOLIS STRING INVERTER

10

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanthapuram



TENAGA ENERGY SOLUTIONS LLP
Reg ofc :845-5 Ground floor MKK Building
Near Al Salama Eye Hospital , Perinthalmanna
Pin 679322; Ph 9544243300, 9846243300
Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

8. 15 KW ONGRID POWER PLANT :CLIENT NAME : MOHAMMED SILIN, MD EUROGRAPH LLC UAE & SOUDI ARABIA; SITEAT HIS RESIDENCE PARAPPANPOYIL, THAMARASSERY;SHEET ROOF AND CUSTOMISED DESIGN ;INVERTER : SOLIS & POLYCAB ; PV MODULE : SAATVIK TOPCON
9. 10KW ONGRID UNDER DBT SUBSIDY SCHEME , MOAHMMED EASSA ,QATAR KMCC AT HIS RESIDENCE MINNAT MAHAL MOODAL KUTTIPURAM, HAVELLS MI COMBO
10. 11KW SOLAR ONGRID PROJECT :CLIENT NAME : Mr. JAMSHEER, AYIRANAZHIPADI, MANKADA;INVERTER: 10KW POLYCAB; PV MODULES : WAAREE SOLAR
11. 10KW ONGRID PLANT AT SKYLINE VILLA , ENRAJIPPALAM KOZHIKODE , MOHAMMED MUSTHAFA
12. 10KW ONGRID PLANT AT MALAPARAMBA , KOZHIKODE Dr SHAMEEM

LIST OF PROJECTS IN KARNATAKA : UNDER MSECOM , KARNATAKA ELECTRICITY BAORD

- 8KW ONGRID PLANT ; LOCATION : MANGALORE, KARNATAKA
CLINET NAME : SHAFEQUE
- 8KW SOLAR PLANT ; CLIENT NAME : BAWA BUILDERS
LOCATION : MANGALORE ; STRING INVERTER : SOLIS
- PV MODULES : VIKRAM; 5KW SOLAR PLANT : MASJID,
MOODBIDRI DAKSHIN KARNATAKA ; STRING INVERTER : SOLIS
PV MODULES : VIKRAM

GOVT PROJECTS:

1. **100MW SOLAR CITY PORJECT IN HIRUVANATHAPURAM UNDER ANERT**
2. **12 KW HYBRID SOLAR PLANT AT FHC, KAKKODI KOZHIKODE UNDER ANERT**

11

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanthapuram



TENAGA ENERGY SOLUTIONS LLP
Reg ofc :845-5 Ground floor MKK Building
Near Al Salama Eye Hospital , Perinthalmanna
Pin 679322; Ph 9544243300, 9846243300
Email : info@tenaga.in, www.tenaga.in

TES/PMNA/GT-PMSURYA/b2c/062025-19

17/06/2025

3. **17 KW ONGRID SOLAR PLANT AT VARIOUS GOVT BUILDINGS IN PASRASSALA, UNDER ANERT**
4. **5KW ongrid plant at Govt Unani center , Mogral Kasaragodunder ANERT**
5. **EMPANELLED UNDER THE 2KW SOLAR PLANTS FOR 1000 LIFE MISSION HOUSEHOLDS UNDER ANERT KERALA NAD MNRE**

12

Branch offices : Thampanoor , Thiruvanthapuram &Uppala, Kasaragod
Empanelment under KSEBL ,ANERT , MNRE ,& Solarcity Thiruvanathapuram

